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MC-OCR Challenge 2021: Deep Learning Approach for Vietnamese Receipts OCR

Doanh C. Bui, Dung Truong, Nguyen D. Vo, Khang Nguyen
University of Information Technology, Ho Chi Minh City, Vietnam
Vietnam National University, Ho Chi Minh City, Vietnam
19521366@gm.uit.edu.vn, {dungtruongq, nguyenvd, khangnttm}@uit.edu.vn

***Abstract*—Receipts OCR has made a significant improvement on accounting, which has attracted much attention of the research community in the field of computer vision as well as natural language processing. In this paper, we solve the problem of extracting pieces of information on Vietnamese receipts including seller, address, timestamp, and total cost. We divided this into two problems: detecting locations of information and using an OCR model to recognize texts. In this paper, we propose a pipeline that employs Faster R-CNN as an information location detector and training a Transformer model for text recognition. Through experiments, we achieved CER 32.19%, which is 9.65% higher than previous method CRNN, while pointing out the remaining statements and challenges of this problem.**

***Index Terms*—OCR; Optical Character Recognition; Transformer**

I. Introduction

Revolution 4.0 put high pressure on making a change in enterprises. Along with the development of technology, document digitization has become a "soft weapon" that helps businesses be easily stored and shared, saving a large number of associated costs. However, in Vietnam, paper receipts are still popularly circulated at small retail stores like coffee shops, essential stores,...Therefore, it is necessary to extract and digitize important information from these documents.

Nowadays, receipts are prevalently used and are easily digitized using smartphones. Some vital information in receipts is of much interest such as "seller", "address", "timestamp", "total cost". There are two stages to extract this information on a receipt image: (1) Detecting regions for each type of information, (2) Recognizing texts using the Optical Character Recognition (OCR) technique. (Figure 2). To extract sufficient information, the detection of information regions needs to be complete and accurate. This stage has a strong impact on the text recognizing stage.

Since 2012, deep learning methods showed outstanding results, which are applied to tackle many problems of image classification, detection, and segmentation[1] in many domains: CCTV images, aerial images, document images etc[2]–[5]. Much recent research has shown very efficient OCR methods on tasks such as name card recognition, license plate recognition, and hand-written text recognition[6]–[8] However, receipts OCR requires extremely high accuracy because it is related to much important information[4]. This problem remains many challenges because the receipt has various formats so that it has difficulty in detecting exactly the locations of needed information. Recognizing texts is also a big problem, many effects cause difficulty on this task but the biggest is the quality of the image.

Figure 1: An example of result

In this paper, we applied Faster R-CNN as an information location detector. About text recognition, we first followed the solution of Huang *et al.* in ICDAR2019, used CRNN architecture to OCR model[4] but this attempt could not help us solve this task optimally for the Vietnamese dataset. Instead, we applied a Transformer model and the result became more positive.

The rest of the paper is structured as follows. Section II provides an overview of related work. Section III describes how we applied it to our research. Section IV shows our experiments and discussions. Finally, conclusions are drawn in Section V.

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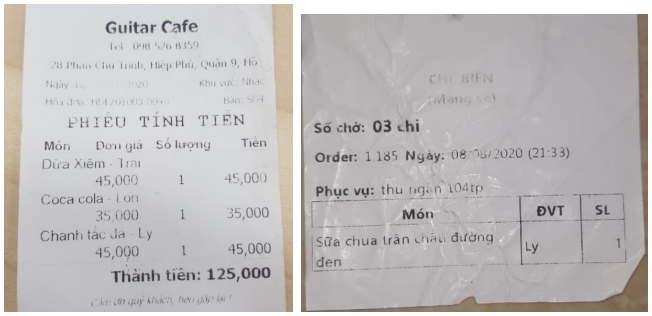

Figure 2: Some challenging receipt images

II. Related Work

In this section, we introduce related works that will be employed in our research.

A. An approach

The basic approach of the OCR problem is that with an input image is printed text or handwriting, we will perform the following steps: scanning, determining the position of the texts, pre-processing, feature extraction, and recognition.[9][10] (Figure 3)

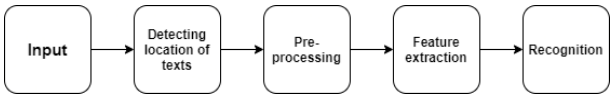

Figure 3: An approach for OCR system

In this paper, we will focus on experiments in two phases: pre-processing and feature extractor.

B. Faster R-CNN

Faster R-CNN[11] is the follow-on work to successful yesterday's Fast R-CNN by Shaoqing Ren et al. The key point of the improvement is:

1) Proposed region proposal network (RPN) that shares full-image convolutional features with detection network (Fast R-CNN). The convolutional computations are shared across the RPN and the Fast R-CNN. This reduces the computational time.
2) Authors introduced novel "anchor" boxes that serve as references in generating region proposals.

The reason is that they observed convolutional feature maps used by region-based detectors like Fast R-CNN can also be used for generating region proposals. Thus, the authors introduced Region Proposal Networks (RPNs) that share convolutional layers with the state-of-the-art object detection network (Fast R-CNN). Thus RPN is a fully convolutional network that simultaneously predicts object bounds and object scores at each position. Now proposal computation is nearly cost-free given the detection network's computation.

Faster R-CNN takes an image with any size as its input. The responsibility of RPN is to generate region proposals. All generated region proposals are converted to a fixed-length feature by using the ROI Pooling layer. Then it will be classified by Fast R-CNN. At the end of the mentioned processes, class scores of the detected objects and their bounding boxes are returned.

There have been related studies utilizing the power of the Faster R-CNN method in object detection problems in documents[12][5]. So, in this research, we applied this method for detecting locations of information.

C. Backbone architectures

1) ResNet: ResNet is a convolutional neural network that is proposed by He *et al.* in 2015. The key point of this work is the definition of the residual block[13].

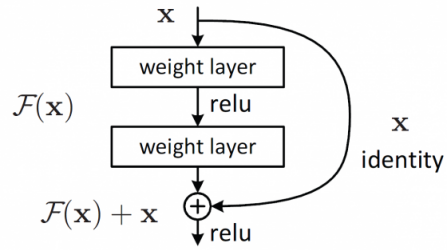

Figure 4: Residual Learning: a building block[13]

As figure 4, the residual block and be understood as identity mapping (Skip connection), this identity mapping is just adding the output from the previous layer to the layer ahead. Thanks to this modification, networks are built from ResNet architecture can efficiently gain accuracy from greatly increased depth.

2) ResNeXt: Introduced by Xie *et al.* in 2017, ResNext is a convolutional neural network that has fewer hyperparameters than ResNet. A new point introduced by the authors is the concept of "cardinality", an additional dimension on top of the width and depth of ResNet[14]. Cardinality defines the size of the set of transformations.

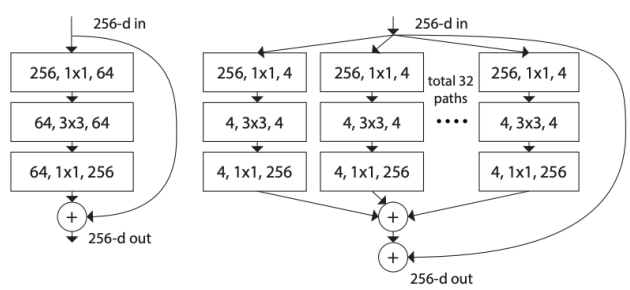

Figure 5: ResNet block (left) and ResNeXt block (right)[14]

Figure 5 shows the difference between ResNet (left) and ResNext (right) block. The ResNext block is shown in figure 5 has a cardinality of 32, which means the same transformations are repeated 32 times and the result is aggregated at the end.

D. Model OCR

1) Convolutional Recurrent Neural Networks: Proposed by Shi *et al.*, Convolutional Recurrent Neural Networks (CRNN)

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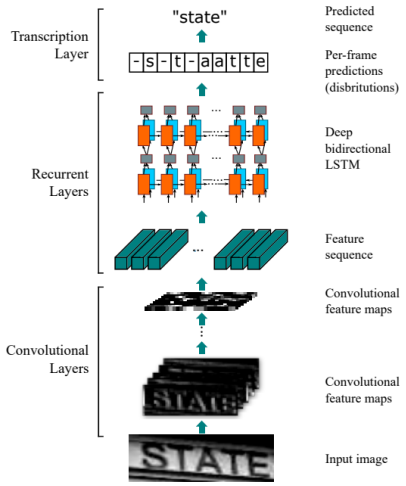


Figure 6: The CRNN architecture[15]

is a solution for Image-based sequence recognition [15]. The network architecture of CRNN consists of three components: convolutional layers, recurrent layers, and a transcription layer, from bottom to top. (Figure 6). The convolutional layers play a role as sequence feature extractor of each input images; a recurrent network on the top is built for making a prediction for each frame of the features extraction, exported by convolution layers; The transcription layer at the top is used for translating each frame predictions by the recurrent layers into a label sequence. But using CRNN, we found that it did not show high efficiency on the Vietnamese receipts dataset. This will be discussed more clearly in Section IV.

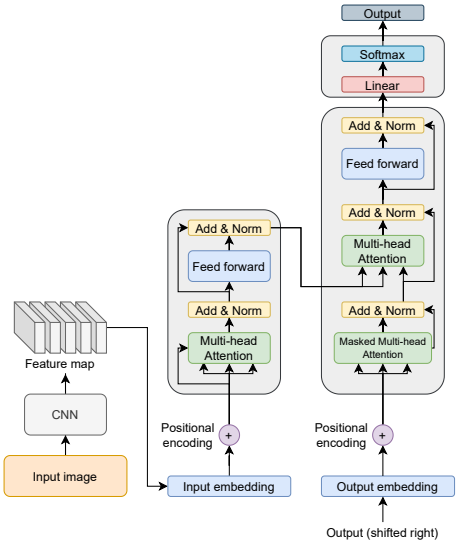
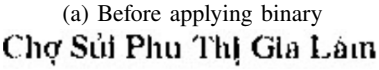


Figure 7: The architecture of Transformer[16]

2) Transformer: First introduced by Vaswani *et al.*, Transformer architecture outperforms many previous recurrent network architectures such as RNN, LSTM.

Like previous recurrent architectures, Transformer is an encoder-decoder architecture mode. The feature extractor

(a) Before applying binary



(b) After applying binary

Figure 8: An example of applying binary on images

firstly produced feature maps of an image, then it will be fit to the encoder. The encoder maps these feature maps $(x_1, x_2, ...x_n)$ to a sequence of continuous representations $(z_1, z_2, ..., z_n)$. The output is then fitted to the decoder for generating symbols $(y_1, y_2, ..., y_n)$ one element at a time, the next symbol will be predicted depending on the previous.

Different from previous regression networks, Vaswani *et al.* propose Scaled Dot-Product Attention to calculate attention weights for the sequences present in the input embedding. Given Q, K, V are the query, key and value matrix, respectively, the attention weight matrix is calculated by Equation 1:

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V \tag{1}$$

With $\frac{1}{\sqrt{d_k}}$ as the scale ratio, it helps to reduce the magnitude of dot products when d_k is too large. Applying dot products to calculate self-attention makes computation faster. The authors also use the multi-head attention mechanism, which helps to learn many attentions simultaneously and then combine them to get the final value via the concatenation operation. This contribution make the Transformer architecture become powerful in sequence-to-sequence problems.

Vaswani *et al.* also propose to add position information of elements in input embedding by using sine and cosine functions of different frequencies (Equation 2, 3):

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}}) \tag{2}$$

$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}}) \tag{3}$$

III. Methodology

In this section, we explain more clearly the method (Figure 10) that we experimented with.

A. Pre-processing techniques

Pre-processing techniques play a significant role to improve the efficiency of the OCR model. We will describe some techniques that we applied in this research as below.

1) Binary image: A binary image is a monochromatic image consisting of pixels that can have one of exactly two colors, usually black and white. In an OCR task, images should be converted from RGB to Binary, it will help the process of training the OCR model achieve high accuracy. Figure 8 shows an example result of this process.

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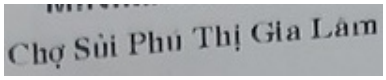
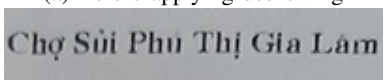


(a) Before applying deskewing



(b) After applying deskewing

Figure 9: An example of applying deskewing on images

2) Projection Profile Method: Because documents images have various angles depend on humans so that images are not always straight as we expect. So deskewing is a significant task on OCR. One of the algorithms that help us solve this problem is Projection profile analysis[17]. The main idea is to rotate the image at various angles and generate a histogram of pixels in each case. For determining the skew angle, we compare the maximum difference between peaks and using this skew angle to rotate the image to correct the skew. Figure 9 is the output of this process.

B. Detecting information locations

To solve the detecting locations problem, we take advantage of the power of current object detection methods. Therefore, the Faster R-CNN method is applied for this task. As Faster R-CNN is both a detector and classifier, the returned outputs are locations and type of its information.

We used ResNeXt-101 architecture with cardinality = 64 and ResNet-101 as backbones to train Faster-RCNN model.

C. Recognizing texts

To solve the recognizing problem, first, we cropped locations of information from the full dataset, texts in cropped images can be skewed, so we recalculated the tilt and straighten the text (deskewing) in the image using the Projection Profile method; converting RGB to Binary is also applied for increasing the efficiency of the OCR model. By cropping and solving images, we created a new dataset just to serve for training OCR. We trained four Transformer models, each model belonged to one information type.

D. Data augmentation

In an attempt to improve efficiency when training Transformer models, we experiment to check the efficiency of models with and without data augmentation[18]. Our augmented data is a mix of original, binary, and deskewed text images.

E. Metrics

1) CER: We use the CER score as a metric for evaluation, which is the same metric as Task 2 in the MC-OCR competition.

First, Levenshtein distance [19] of all fields is computed. Then, the normalized score of the Levenshtein distance of all key information is calculated as the final score. The CER score is represented by the Equation 4.

$$CER = \frac{1}{L}\sum_{1}^{N}(i+s+d). \tag{4}$$

where $\sum_1^N(l_i)$ is the total length of all reference texts of the test set; l_i is the length of ith document; N is the total number of text samples; $(i+s+d)$ corresponds to the minimal numbers of character insertions i, substitutions s, and deletions d required to transform the reference text into the OCR output.

2) WER: In the parallel of computing CER as a metric, we also use Word Error Rate (WER) (Equation 5), which is the same CER metric but calculated on word unit.

$$WER = \frac{1}{L_w}\sum_{1}^{N}(i_w+s_w+d_w). \tag{5}$$

where L_w is the number of words in the reference texts, s_w is the number of words substituted, d_w the number of words deleted, and i_w the number of words inserted required to transform the output texts into the reference texts.

IV. Evaluation and Discussion

A. Dataset

In this study, we did experiments on a public dataset published in the "RIVF2021 MC-OCR" competition[1]. We only used a dataset that annotations are available [20]. As mentioned in section III, we split the dataset into 5 folds and did cross-validation. The train/test ratio is 80:20. The dataset includes 1149 images of receipts contain 6550 bounding boxes that are annotated in 4 classes: seller, address, timestamp, total cost; each image also has texts as ground truth. Figure 11 shows the distribution of information types of dataset.

B. Result

Since the results of text recognition are depended on the number of bounding boxes which Faster R-CNN model returns, we did a small experiment to observe the affection of confidence score (how accurate it thinks the box is that predicts) (Figure 12), we also evaluate the influence of data augmentation.

As a result, it was shown that the higher the confidence score, the better the results. The reason is that the lower confidence score will return more bounding boxes and maybe they do not include expected text, causing difficulties in text recognition. Besides, the data augmentation has also brought about a marked effect.

Our experiments in Table 3 mainly focus on the impact of pre-processing techniques as well as the difference between the CRNN and Transformer model. We evaluate these on 02 backbones ResNet-101 and ResNeXt-101 64x4d mentioned in Section III.

The result proves that our proposed workflow using Transformer is significantly different from using CRNN. Besides, the combination of pre-processing techniques also shows diverse CER scores with applying both Binary and Deskewing

[1] https://competitions.codalab.org/competitions/27798

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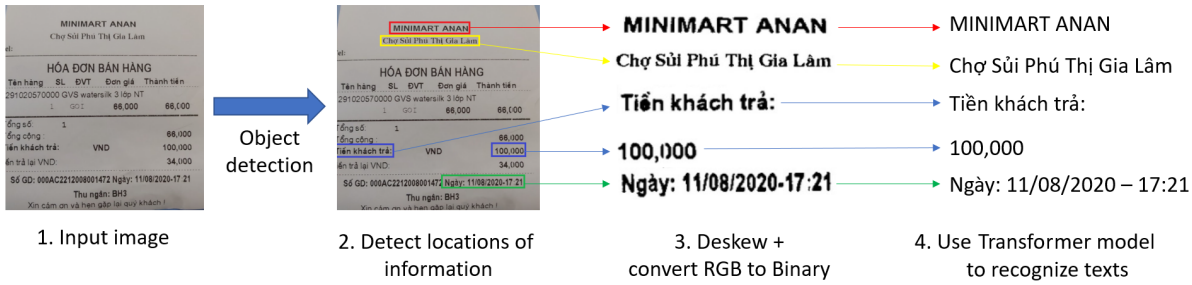

Figure 10: Diagram depicts the steps of our method. The locations of information on receipt will be predicted by the object detection model. These detections then will be deskewed and converted to binary. Finally, we will apply a specific Transformer model for each class to receive the final result.

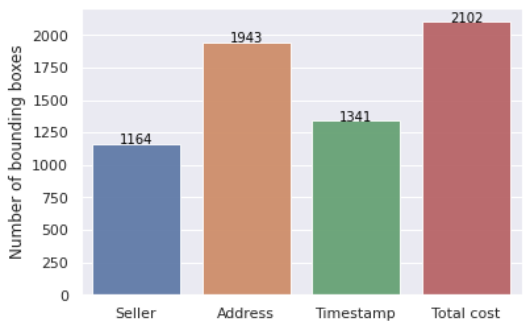

Figure 11: Distribution of information types of dataset

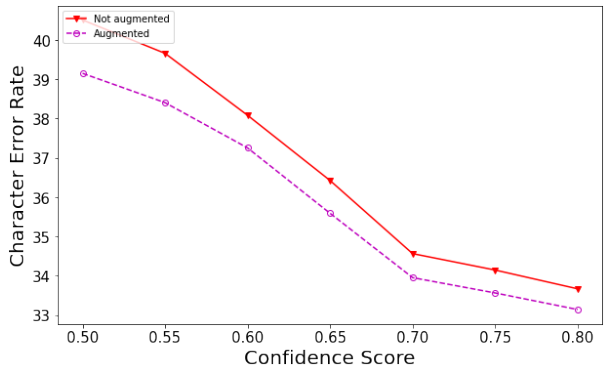

Figure 12: The effectiveness of confidence score and with and without data augmentation

shows the highest efficiency. Data augmentation is also an important factor of the enhancement, the CER score improves 0.66% in ResNet-101 and 0.53% in ResNeXt-101 64x4d when applying this. The final thing in our experiment discovered that is ResNet-101 is more suitable for receipts OCR problem than ResNeXt-101 64x4d, the best result in Table I is **CER 32.19%** using ResNet-101, 0.95% higher than ResNeXt-101 64x4d. In both detection and text recognition tasks, we used 5 folds cross-validation to ensure the effectiveness of the proposed solution.

V. Conclusion And Future Work

In this paper, we indicated an approach for the receipts OCR, analyzed some related works, and applied them to propose the solution for this problem. We divided this problem into 2 tasks: detection locations of information on receipts and text recognition. In the detection task, we used Faster-RCNN and did experiments on 2 backbones: ResNet-101 and ResNeXt-101 64x4d. In the text recognition task, we trained a Transformer model with and without text data augmentation. The best result in our experiment is CER 32.19%, using ResNet-101 and Transformer model trained on augmented text data. In the future, we will try to test multiple methods of object detection and combine them for better results on the detection task as well as enhancing the Transformer model.

Acknowledgment

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Table I: Text recognition model evalutaion

Backbone	Pre-processing		Model OCR		Data aug	CER (%)	WER (%)
	Binary	Deskew	CRNN	Transformer			
ResNet-101			✓			43.55	79
	✓		✓			42.25	71
		✓	✓			43.45	78
	✓	✓	✓			41.84	70
				✓		32.7	44.48
	✓			✓		32.44	43.33
		✓		✓		32.47	43.29
	✓	✓		✓		32.85	44
	✓	✓		✓	✓	**32.19**	**43.29**
ResNeXt-101 64x4d			✓			43.34	78
	✓		✓			43.32	72.43
		✓	✓			43.57	78.6
	✓	✓	✓			42.88	71.83
				✓		33.84	45.78
	✓			✓		33.45	44.68
		✓		✓		33.17	43.85
	✓	✓		✓		33.14	44
	✓	✓		✓	✓	33.67	45.8

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